

A Guide to Statistical Network Modeling

Session 1.1: Foundations of Static Exponential-Family Random Graph Models

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instats Seminar

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Companion computations and reproducibility

- **Public repository:** github.com/mboudour/no-code-statistical-network-modeling.
- **Companion Streamlit app hosted on Render:** no-code-statistical-network-modeling.onrender.com; open **Five worked public networks** for the five guided public-data workflows.
- **No-code workflow:** select a documented public network, inspect its support and covariates, specify a static ERGM, fit it with the standard statnet engine, and preserve the output record.
- **Scope of this session:** binary *static* graph distributions; curved terms, degeneracy-aware refinement, and the full standard ERGM diagnostic audit are developed in Session 1.2, while temporal models are developed later.
- **Principle:** an application can make computations reproducible; it cannot determine a credible network boundary, causal interpretation, or model specification automatically.

Learning objectives

By the end of this session, we will be able to:

- define a network observation, its **support**, and its **risk set**;
- distinguish **edge independence**, **dyad independence**, and structural dependence;
- state the canonical static **exponential-family random graph model** (ERGM);
- use **change statistics** to interpret conditional log-odds;
- translate substantive network questions into a modest, documented ERGM specification; and
- identify claims that an ERGM output does *not* establish.

Session route

- ① **Data and support:** what graph is observed, and which dyads are possible?
- ② **Dependence:** why independent-dyad regression is often inadequate.
- ③ **Static ERGM:** probability law, statistics, parameters, and conditional interpretation.
- ④ **Specification:** structural and covariate statistics chosen from a research question.
- ⑤ **Computation:** likelihood, simulation, pseudolikelihood, and five public-network workflows.

Why statistical network modeling?

- Conventional regression treats observations as independent conditional on covariates; a **network tie** can depend on other ties.
- **Reciprocity:** Y_{ij} may be informative about Y_{ji} .
- **Closure:** a tie from i to j can be associated with paths through third actors.
- **Degree heterogeneity:** ties sharing an actor can reflect common activity or popularity processes.
- A statistical network model therefore defines a probability distribution for an **entire graph**, not an unrelated collection of rows.

Network data: actors, relations, and graphs

- Let $V = \{1, \dots, n\}$ be the **actor set**; actors can be people, organizations, places, species, or events.
- Let Y_{ij} record the relation from i to j ; for a binary tie, $Y_{ij} \in \{0, 1\}$.
- The adjacency matrix $Y = (Y_{ij})$ is random; the observed network is its realization y .
- The **unit of probabilistic modeling** is Y , while an actor, dyad, or local configuration can enter a statistic of Y .
- A network question must state the relation, observation window, actor boundary, and whether a tie is directed or valued.

Directed, undirected, and loopless support

- In a **directed** network, Y_{ij} and Y_{ji} are distinct potential ties.
- In an **undirected** network, $Y_{ij} = Y_{ji}$ and an unordered pair $\{i, j\}$ has one edge variable.
- A graph is **loopless** when $Y_{ii} = 0$; this is common but must be stated rather than assumed universally.
- For a directed loopless binary graph, there are $n(n - 1)$ potential ordered dyads.
- For an undirected loopless binary graph, there are $\binom{n}{2} = n(n - 1)/2$ potential unordered dyads.

One-mode and two-mode networks

- A **one-mode** network has ties among actors in one actor set, such as firms linked to firms.
- A **two-mode** or **bipartite** network has two disjoint modes, such as women connected to events.
- Bipartite support admits ties only across modes; within-mode dyads are **structural zeros**, not observed absences.
- Projecting a bipartite graph to one mode changes the data object, induces dependence, and must not be treated as an innocuous formatting step.
- An ERGM specification must use statistics and support appropriate to the network mode structure.

Static is an observation design, not a claim of no change

- A **static ERGM** models the distribution of a network observed at one designated time or aggregation window.
- The network may have evolved before observation; a static model does not identify its unobserved sequence of tie changes.
- Aggregating contacts over a long window can convert repeated events into a binary “ever tied” relation and can conceal duration or ordering.
- A static interpretation is appropriate only for a question about the graph distribution at the stated observation design.
- Network evolution is treated with TERGMs, STERGMs, and SAOMs in later sessions.

The admissible dyad set and graph support

For a loopless graph, define the admissible dyad set

$$D = \begin{cases} \{(i, j) : i \neq j\}, & \text{directed,} \\ \{(i, j) : i < j\}, & \text{undirected.} \end{cases}$$

- The **sample space** $\mathcal{Y}$ is the set of all graph configurations allowed by the design.
- With no additional constraints, a binary undirected graph has $|\mathcal{Y}| = 2^{\binom{n}{2}}$ configurations.
- Structural constraints, bipartite node membership, and forbidden relations alter $\mathcal{Y}$ and therefore alter the probability model.

Network boundary and risk set

- The **boundary** specifies which actors are included in V .
- The **risk set** specifies which dyads could in principle form or be observed during the study design.
- Excluding actors who carry relational traffic can distort observed degree, mixing, and closure configurations.
- An ERGM parameter is conditional on the selected actor set and support; it is not automatically portable to an unobserved wider network.
- Report how actors entered the network and which possible dyads were excluded by design.

Observed zero, missing tie, and structural zero

- An **observed zero** is an admissible dyad recorded as absent.
- A **missing tie** is an admissible dyad whose state is unobserved; it requires an observation-process assumption or an observed-data treatment.
- A **structural zero** is a dyad not in the support, such as a within-mode pair in a bipartite network.
- Coding missing dyads as zeros can manufacture sparsity and bias structural statistics.
- Removing structural zeros after fitting is also incorrect: support must be specified before the likelihood is defined.

Actor and dyadic covariates

- An **actor covariate** x_i is measured on an actor: role, seniority, wealth, group, or behavior.
- A **dyadic covariate** x_{ij} is measured on a pair: distance, co-membership, exposure, or opportunity.
- Covariates must be aligned to the actor identities and observation design that define Y .
- A covariate measured after the observed relation can be useful descriptively but may be unsuitable as a causal predictor.
- ERGMs represent covariates through specified network statistics rather than through a generic row-level regression design matrix.

Descriptive statistics versus generative parameters

- A **descriptive statistic** summarizes the observed graph, for example edge count or clustering coefficient.
- A **generative statistic** $g_k(Y, X)$ enters a probability model for many possible graphs in $\mathcal{Y}$.
- A triangle count observed in y does not by itself imply a positive ERGM closure parameter.
- A homophilous observed mixing pattern can reflect group composition, degree heterogeneity, reciprocity, or other modeled features.
- ERGM interpretation is therefore **conditional on the full specification**, not a one-statistic narrative.

Relational dependence

- **Relational dependence** means that the distribution of one tie can depend on other ties after covariates are considered.
- Dependence is not an unspecified residual-correlation adjustment; it is represented through declared graph configurations.
- A reciprocity statistic specifies a within-dyad dependence mechanism.
- A closure statistic specifies dependence involving triples or longer local configurations.
- The substantive task is to choose statistics that encode a plausible relational process while avoiding indiscriminate term accumulation.

Edge independence and dyad independence

- **Edge independence** assumes all admissible edge variables are mutually independent conditional on covariates.
- In an undirected binary graph, edge independence and dyad independence coincide because each unordered dyad contains one edge variable.
- In a directed graph, **dyad independence** allows (Y_{ij}, Y_{ji}) to be dependent while distinct unordered dyads remain independent.
- A reciprocity ERGM can be dyad independent but not edge independent.
- This distinction determines which factorization and conditional interpretation are available.

A local-dependence illustration

- Under the Frank–Strauss Markov condition for an undirected graph, disjoint-edge variables satisfy

$$Y_{ij} \perp Y_{kl} \mid Y_{-(ij),(kl)} \quad \text{when } \{i, j\} \cap \{k, l\} = \emptyset.$$

- A positive distribution satisfying this condition can be represented using edge, k -star, and triangle configurations.
- General ERGMs need not impose this particular Markov restriction; they may represent broader or nonlocal dependence.
- The selected sufficient statistics, not the acronym “ERGM,” carry the substantive dependence assumptions.

The canonical static ERGM

$$\Pr_{\boldsymbol{\theta}}(Y = y \mid X) = \frac{\exp\{\boldsymbol{\theta}^T g(y, X)\}}{\kappa(\boldsymbol{\theta}, X)}, \quad \kappa(\boldsymbol{\theta}, X) = \sum_{y' \in \mathcal{Y}} \exp\{\boldsymbol{\theta}^T g(y', X)\}.$$

- $g(y, X) = (g_1, \dots, g_p)^T$ is the vector of specified **network statistics**.
- $\boldsymbol{\theta} \in \mathbb{R}^p$ is the canonical parameter vector.
- κ is the **normalizing constant**; it assigns total probability one over the selected support.

Why the normalizing constant matters

- $\kappa(\theta, X)$ sums over every graph allowed by $\mathcal{Y}$.
- Its size grows exponentially in the number of possible dyads; direct enumeration is infeasible except for very small graphs.
- The intractable normalizing constant motivates simulation-based estimation and simulation-based diagnostics.
- It does *not* disappear from a likelihood merely because the observed graph is represented by an adjacency matrix.
- Some comparative expressions are tractable because κ cancels, as in graph probability ratios and change-statistic interpretations.

Exponential-family identities

Let $\psi(\boldsymbol{\theta}; X) = \log \kappa(\boldsymbol{\theta}, X)$. Then, for the canonical family,

$$\nabla_{\boldsymbol{\theta}} \psi(\boldsymbol{\theta}; X) = \mathbb{E}_{\boldsymbol{\theta}}\{g(Y, X) \mid X\}, \quad \nabla_{\boldsymbol{\theta}}^2 \psi(\boldsymbol{\theta}; X) = \text{Var}_{\boldsymbol{\theta}}\{g(Y, X) \mid X\}.$$

- The likelihood score compares observed sufficient statistics with their model expectations.
- The second identity links curvature with variability of the statistics and expected Fisher information.
- These canonical identities require modification for curved parameterizations, treated in Session 1.2.

Probability ratios compare complete graphs

For $y_a, y_b \in \mathcal{Y}$ under the same covariates,

$$\frac{\Pr_{\theta}(Y = y_a \mid X)}{\Pr_{\theta}(Y = y_b \mid X)} = \exp\left[\boldsymbol{\theta}^{\top}\{g(y_a, X) - g(y_b, X)\}\right].$$

- The normalizing constant cancels in the ratio.
- Parameters assign relative weight to differences in network statistics.
- This is a comparison of **entire network configurations**, not a marginal probability for a randomly chosen tie.
- A one-dyad toggle yields the conditional-logit interpretation on the next slides.

Change statistics

For an admissible binary dyad (i, j) , let y_{ij}^+ and y_{ij}^- agree except that the dyad is set to one or zero. Define

$$\Delta_{ij}g(y, X) = g(y_{ij}^+, X) - g(y_{ij}^-, X).$$

- A **change statistic** records how each modeled feature would change if one tie were toggled while all other ties are held fixed.
- It depends on y_{-ij} and X , but not on the current value of Y_{ij} .
- Change statistics make local structural context explicit: the same potential tie can have different model contribution in different network neighborhoods.

Conditional-logit identity

$$\text{logit}_{\theta} \Pr(Y_{ij} = 1 \mid Y_{-ij} = y_{-ij}, X) = \theta^T \Delta_{ij} g(y, X).$$

- Each coefficient weights the corresponding **change statistic** in the conditional log-odds of a tie.
- “Conditional” means conditional on the complete remainder of the graph and included covariates.
- For a toggle with $\Delta_{ij} g_k = 1$, exponentiating θ_k gives the multiplicative contribution of that term to the conditional odds, holding the other change statistics fixed.
- This is not a population-average tie probability and is not automatically a causal effect.

Toggle interpretation requires support

- The conditional-logit identity applies when both y_{ij}^+ and y_{ij}^- belong to $\mathcal{Y}$.
- A structural zero has no binary toggle comparison because its tie-present state is not admissible.
- Constrained supports can restrict which single-edge toggles are valid; for example, fixed-degree support generally requires degree-preserving moves rather than arbitrary single-edge toggles.
- Bipartite support permits toggles only across modes.
- Before interpreting a coefficient, verify that the corresponding tie comparison is defined under the network's stated support.
- Software options for directedness, bipartite structure, missing ties, and constraints are statistical assumptions, not merely formatting controls.

The edges statistic: baseline sparsity

- For a binary graph, $g_{\text{edges}}(y) = \sum_{(i,j) \in D} y_{ij}$ counts present ties.
- Its coefficient controls baseline conditional tie propensity after all other specified effects are included.
- A negative edges coefficient is common in sparse networks, but its magnitude depends on the support and the rest of the model.
- It is not simply the observed density transformed by a logistic function once structural and covariate terms are included.
- Every introductory ERGM needs a clearly interpreted baseline relation term.

Reciprocity in a directed network

$$g_{\text{mutual}}(y) = \sum_{i < j} y_{ij} y_{ji}.$$

- The statistic counts mutually tied unordered pairs in a directed graph.
- Its change statistic is one when toggling $i \rightarrow j$ completes a reciprocal dyad, and zero otherwise.
- A positive coefficient favors reciprocated dyads *relative to* otherwise comparable directed graphs under the full model.
- Reciprocity is not defined for an undirected graph and must not be added mechanically to a one-mode undirected or bipartite specification.

Degree patterns and centralization

- The **degree** of an undirected actor counts incident ties; directed graphs distinguish indegree and outdegree.
- A degree distribution is descriptive; it can reveal isolates, hubs, and dispersion that a simple edges-only model will not reproduce.
- Star statistics count configurations sharing a focal actor and can represent degree-related structure.
- Raw high degree is not evidence that a particular star parameter belongs in a model; the statistic must answer a substantive question and be diagnostically stable.
- Curved geometrically weighted degree terms are deferred to Session 1.2.

Closure and triadic structure

- A triangle in an undirected graph is a fully connected triple; directed closure has several distinct configurations.
- Closure terms can represent a tendency for ties to occur in locally connected neighborhoods.
- Triangle counts overlap strongly: a single tie can complete several triangles, so the change statistic depends on shared partners.
- A positive triangle-related parameter is conditional on the other statistics; it is not proof of a social mechanism such as trust or contagion.
- Session 1.2 treats stable curved alternatives to naïve high-order structural terms.

Actor-level covariate effects

For a scalar attribute x_i in a directed graph, possible statistics include

$$g_{\text{send}}(y, x) = \sum_{i \neq j} y_{ij} x_i, \quad g_{\text{recv}}(y, x) = \sum_{i \neq j} y_{ij} x_j.$$

- A **sender** effect asks whether actors with larger x_i tend to send more ties.
- A **receiver** effect asks whether actors with larger x_j tend to receive more ties.
- These are distinct hypotheses; a single correlation between degree and an attribute cannot distinguish them.
- Interpret both conditionally on structural and other covariate terms included in the model.

Homophily and mixing

For a categorical attribute x_i , an undirected matching statistic is

$$g_{\text{match}}(y, x) = \sum_{i < j} y_{ij} \mathbf{1}(x_i = x_j).$$

- A positive matching coefficient favors same-category ties relative to cross-category ties after other terms are held fixed.
- A **mixing matrix** represents ties through category-pair statistics, allowing category-specific patterns that a single matching statistic can conceal.
- Directed and undirected mixing statistics must be defined separately; an undirected category-pair representation must not double-count dyads.
- Homophily coefficients do not automatically identify preference: opportunity, group size, unmeasured similarity, and boundary design may contribute.

A coefficient is conditional, not causal

- ERGM parameters describe relative probability contributions conditional on the rest of the graph, selected support, covariates, and model terms.
- A positive covariate or homophily effect is not by itself an intervention effect.
- A positive closure term is not direct evidence of a psychological preference for triangles.
- Causal claims require a design, temporal ordering, treatment definition, and explicit account of confounding and interference.
- Scientific interpretation should separate **model-based association** from a stronger causal explanation.

From research question to ERGM specification

- ➊ Define actors, relation, time window, boundary, and support.
- ➋ State a graph-level question: reciprocity, mixing, closure, degree concentration, or covariate-associated ties.
- ➌ Select a small set of statistics justified by the question and network type.
- ➍ Decide which covariates are pre-relation, observed, and aligned to actors or dyads.
- ➎ Fit, inspect computational diagnostics, compare simulated and observed features, and revise only with a documented rationale.

Likelihood and the estimation problem

- For observed graph y , the log likelihood is

$$\ell(\boldsymbol{\theta}) = \boldsymbol{\theta}^\top g(y, X) - \log \kappa(\boldsymbol{\theta}, X).$$

- The score equates observed statistics with their model expectations.
- Direct maximum likelihood requires expectations under a distribution with an intractable normalizing constant.
- **Monte Carlo maximum likelihood estimation** approximates those expectations by simulating networks from trial parameter values.
- Convergence, sample-space behavior, and simulation quality are part of the statistical result, not post-processing details.

MCMC maximum likelihood: conceptual cycle

- ➊ Start from a parameter value and simulate networks by admissible tie toggles or other valid moves.
- ➋ Estimate expected sufficient statistics under the current model.
- ➌ Compare simulation expectations with $g(y, X)$ and update the parameter value.
- ➍ Repeat until the estimating discrepancy is sufficiently small under stated convergence criteria.
- ➎ Retain settings, random seed policy, warnings, and diagnostics in the reproducibility record.

Maximum pseudolikelihood: useful but limited

- **Maximum pseudolikelihood estimation** (MPLE) forms a pseudolikelihood by multiplying the full conditional probabilities of individual dyads, rather than evaluating the joint ERGM likelihood.
- It is computationally convenient and can provide a starting value for MCMC estimation.
- Under dependence, this product of full conditionals is not the joint ERGM likelihood; standard errors and point estimates can be biased or overconfident relative to the likelihood target.
- A logistic-regression-looking output is not evidence that dependence has disappeared.
- The app labels pseudolikelihood as an approximation; reported primary results use the standard simulation-based ERGM engine when it converges.

Uncertainty and inferential discipline

- A standard error quantifies uncertainty conditional on the fitted model, data boundary, support, and estimation procedure.
- A confidence interval does not repair an implausible statistic, a poor network boundary, missing data, or failed MCMC convergence.
- Model selection based only on coefficient significance risks specification search and selective reporting.
- Report estimates, uncertainty, simulation settings, diagnostics, and the scope of the population to which the observed network pertains.
- Avoid dichotomous conclusions such as “proved a network mechanism.”

ERGM diagnostic audit

- **Estimation behavior** and **goodness of fit** are different: convergence concerns the fitted calculation, whereas goodness of fit concerns networks simulated from the fitted distribution.
- Preserve settings, messages, and unresolved concerns; Session 1.2 conducts the full support-appropriate MCMC and simulation-based goodness-of-fit audit.

Five worked public networks in the companion app

Network	Network type and introductory workflow	Static support
Florentine families	Business relations; edges and documented family attributes	Undirected, binary, one-mode
Sampson's monastery	Interpersonal nominations; edges, reciprocity, group mixing	Directed, binary, one-mode
Lazega law firm	Professional advice; edges, reciprocity, office or status covariates	Directed, binary, one-mode
Southern Women	Woman–event participation; bipartite edges and mode structure	Undirected, binary, two-mode
Kapferer tailor shop	Frequent sociational relations; density and documented closure comparison	Undirected, binary, one-mode

Each example is accompanied by source, data dictionary, validity warning, and reproducible computation record.

Worked example: Florentine families

- **Question:** are business ties associated with documented family attributes after accounting for baseline density?
- **Data object:** one static, undirected, binary family business network with actor attributes.
- **Foundational specification:** an edges term plus a pre-specified attribute term or matching statistic that is meaningful for the recorded families.
- **Computation:** inspect estimation diagnostics and conditional coefficient estimates, then compare substantively relevant observed network features with their distributions under simulated networks.
- **Limit:** the small historical network supports a focused illustration, not a general causal claim about wealth or political influence.

Worked example: Sampson's monastery

- **Question:** do interpersonal nominations exhibit reciprocity and within-group mixing?
- **Data object:** a directed, binary, one-mode sociometric network among monks with recorded group affiliation.
- **Foundational specification:** edges + mutual + group matching, with terms included because each maps to a separate question.
- **Interpretation:** the mutual coefficient concerns conditional odds of reciprocated nominations, not a marginal probability of friendship.
- **Limit:** a single observation wave cannot establish the sequence by which affiliations and nominations co-developed.

Worked example: Lazega law firm

- **Question:** how are professional advice ties conditionally associated with reciprocity and documented organizational attributes?
- **Data object:** a directed, binary, one-mode advice network with actor-level law-firm variables.
- **Foundational specification:** start with edges and reciprocity, then add one documented attribute hypothesis rather than all available covariates.
- **Computation:** retain the network boundary, directedness, and attribute coding in the app's analysis record.
- **Limit:** a cross-sectional model describes the observed advice-network distribution; it does not identify influence or organizational change.

Worked example: Davis Southern Women

- **Question:** how does an affiliation graph differ from a one-mode social graph?
- **Data object:** women linked to social events; a static, binary, bipartite incidence network.
- **Support:** only woman–event ties are admissible; woman–woman and event–event cells are structural zeros in the original graph.
- **Computation:** inspect mode sizes and degree distributions before fitting a bipartite-appropriate baseline ERGM.
- **Limit:** a one-mode projection creates a different network and requires a separately justified analysis.

Worked example: Kapferer tailor shop

- **Question:** do relatively frequent sociational relations display local closure after the baseline density is represented?
- **Data object:** a static, undirected, binary, one-mode network of 39 tailor-shop workers from the first collection period.
- **Foundational specification:** begin with an edges-only model; treat a closure extension as a documented later comparison, not an automatic term.
- **Computation:** inspect support, density, degree distribution, MCMC behavior, and simulated graph summaries before interpreting an added structural effect.
- **Limit:** the 39-worker object is a selected analytic panel and records thresholded frequent interaction, not all social contact in a probability sample of factories.

No-code Session 1.1 computation workflow

- 1 Select one of the five public networks and read its source, network-type, and scope note.
- 2 Inspect actor count, admissible dyads, density, directionality, missingness, attributes, and degree distribution.
- 3 Choose a small supported formula from the session's available terms; the app prevents directed-only terms in undirected data.
- 4 Run the standard `statnet` ERGM computation; examine the coefficient table, MCMC messages, and model limitations before the full diagnostic audit in Session 1.2.
- 5 Download a Markdown computation record containing data source, support, formula, software settings, results, warnings, and limitations.

BYOD: required upload contract

- **Edge table:** source ID and target ID for observed binary ties; a valued relation is not silently dichotomized and belongs in an explicitly justified later workflow.
- **Node table:** one unique actor ID per row and optional pre-specified actor attributes.
- **Design choices:** directed versus undirected; one-mode versus bipartite; actor boundary; structural zeros; missingness; and time window.
- **Minimum checks:** duplicate dyads, self-loops, unmatched node IDs, empty attributes, implausible density, and unsupported formula terms.
- **Principle:** upload does not make an ERGM appropriate; the research question and support must justify the model.

Common invalid or misleading uses

- Fitting an undirected model to a directed relation after discarding direction without a substantive reason.
- Treating unobserved dyads as zeros or ignoring an actor boundary that omits important relational alternatives.
- Adding every available structural statistic or covariate merely because software accepts the formula.
- Interpreting a coefficient as a marginal probability, an intervention effect, or a proof of a social mechanism.
- Reporting a pseudolikelihood result as if it had resolved the ERGM normalizing-constant problem.

A reporting checklist for a static ERGM

- **Data:** actors, relation, observation window, boundary, directionality, mode structure, missingness, and support constraints.
- **Question:** which structural or covariate-associated graph feature is under investigation?
- **Specification:** exact sufficient statistics, covariate coding, and rationale for each term.
- **Estimation:** software, package version, algorithm, simulation controls, convergence status, and warnings.
- **Results:** conditional coefficient interpretation, uncertainty, simulated comparisons, limitations, and scope of inference.

Session 1.1 synthesis

- A static ERGM is a probability model for a graph on an explicitly defined support.
- Its parameters weight **network statistics**; they are interpreted through conditional changes under the rest of the graph.
- Data definition, boundaries, structural zeros, and covariate timing are part of the statistical model.
- For most nontrivial dependent ERGMs, simulation-based methods such as MCMC are used because the normalizing constant cannot be evaluated directly.
- Next: **curved ERGMs, stable structural terms, fit, and degeneracy** in Session 1.2.

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